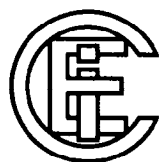


NORME INTERNATIONALE INTERNATIONAL STANDARD



Commission Electrotechnique Internationale

International Electrotechnical Commission

Международная Электротехническая Комиссия

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1987

Transformateurs de puissance

Troisième partie: Niveaux d'isolement et essais diélectriques
Distances d'isolement dans l'air

Power transformers

Part 3: Insulation levels and dielectric tests
External clearances in air

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

POWER TRANSFORMERS**Part 3: Insulation levels and dielectric tests****External clearances in air**

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendation and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This standard has been prepared by IEC Technical Committee No. 14: Power Transformers.

The text of this standard is based on the following documents:

Six Months' Rule	Report on Voting
14(CO)59 14(CO)59A	14(CO)62

Further information can be found in the Report on Voting indicated in the table above.

The following IEC publications are quoted in this standard:

- Publications Nos. 60-2 (1973): High-voltage Test Techniques, Part 2: Test Procedures.
71-1 (1976): Insulation Co-ordination, Part 1: Terms, Definitions, Principles and Rules.
71-3 (1982): Part 3: Phase-to-phase Insulation Co-ordination. Principles, Rules and Application Guide.
76-3 (1980): Power Transformers, Part 3: Insulation Levels and Dielectric Tests.
137 (1984): Bushings for Alternating Voltages above 1 000 V.

POWER TRANSFORMERS

Part 3: Insulation levels and dielectric tests

External clearances in air

1. Introduction

This standard specifies clearances in air between live parts of bushings on oil-immersed power transformers and to objects at earth potential.

Clearances in air are understood to mean, in this standard, distances where the electrostatic field is free from disturbance by insulator bodies. The standard does not deal with the requirements of effective flashover distance or creepage distance along the bushing insulators nor does it consider the risk from intrusion of birds or animals.

The insulation levels and dielectric tests which are specified in IEC Publication 76-3: Power Transformers, Part 3: Insulation Levels and Dielectric Tests, apply, for oil-immersed transformers, to the internal insulation only (see beginning of Clause 2 of that publication).

It is reasonable that the rated withstand voltage values which are specified for the internal insulation of the transformer should also be taken as a reference for its external insulation. This may, however, not be true in all cases. A failure of the non-self-restoring internal insulation is catastrophic and normally leads to the transformer being out of service for a long period, while an external flashover may involve only a short interruption of service without causing lasting damage. Therefore it may be that, for increased safety, higher test voltages are specified by the user for the internal insulation of the transformer than for the external insulation of other components in the system. If such a distinction is made, the external clearances on the transformer should be referred instead to those specified for external insulation withstand voltages.

When establishing the requirements of the present standard in the higher voltage ranges, it has been recognized that the bushing ends normally have a rounded electrode shape. The clearance requirements are valid between such rounded electrodes. It is assumed that conductor clamps with their associated shield electrodes are suitably shaped so that they do not reduce the flashover voltage. It is also assumed that the arrangement of incoming conductors does not reduce the effective clearances provided by the transformer itself.

Note. – If the user intends to make his connection in a particular way which is likely to reduce the effective clearances, this should be mentioned in the enquiry.

In general, the provision of adequate clearances in air becomes technically difficult mainly at high system voltages, particularly for relatively small units, or when the installation space is restricted. The principle followed in the recommendations of this standard is to provide ample, non-critical clearances which are satisfactory without further discussion or proof under various system conditions and in different climates. Where extensive previous experience has indicated that smaller clearances are adequate, there is obviously no need to change.

2. General

The recommended clearances are referred to the rated withstand voltage values of the internal insulation of the transformer, unless otherwise specified in the enquiry and order. When the clearances of the transformer are equal to or larger than the values specified in this standard and the bushings have properly selected ratings according to IEC Publication 137: Bushings for Alternating Voltages above 1000 V, then the external insulation of the transformer shall be regarded as satisfactory without further testing.

Notes 1. – The impulse withstand strength of the external insulation is polarity dependent, in contrast to what is assumed for the internal insulation. The tests prescribed for the internal insulation of the transformer do not automatically verify that the external insulation is satisfactory. The recommended clearances are dimensioned for the more onerous polarity (positive).

2. – It is recognized that in some cases good experience with well documented national practice allows the continuing use of established smaller clearances without further discussion or testing.

If a clearance smaller than that according to the paragraph above has been used for a contract, a type test may be required on an arrangement simulating the actual clearance, or on the transformer itself. Recommended test procedures for such cases are given.

If the transformer is specified for operation at an altitude higher than 1000 m, the clearance requirements have to be increased by 1% for every 100 m by which the altitude exceeds 1000 m.

Requirements are given for the following clearances:

- clearance phase-to-earth and phase-to-neutral;
- clearance phase-to-phase between phases of the same winding;
- clearance between a line terminal of the high voltage winding and a line terminal of a lower voltage winding.

It follows from the above that the recommended values are in effect minimum values. The design clearances shall be stated on the outline drawing. These are nominal values subject to normal manufacturing tolerances and they have to be selected so that the actual clearances will be at least equal to the specified values. These statements shall be taken as proof that the transformer complies with the recommendations of the standard, or with the modified values which may have been agreed for the particular contract.

3. Bushing clearance requirements as determined by transformer insulation withstand voltages

The requirements are formulated as described below, depending on the U_m voltage value of the winding, and on the method of specifying the transformer insulation.

IEC Publication 71-1: Insulation Co-ordination, Part 1: Terms, Definitions, Principles and Rules, deals with phase-to-earth insulation co-ordination. IEC Publication 71-3: Part 3: Phase-to-phase Insulation Co-ordination – Principles, Rules and Application Guide, deals with phase-to-phase insulation co-ordination. IEC Publications 71-1 and 71-3 make use of different reference withstand voltages for different ranges of U_m voltages. In Ranges A and B ($U_m < 300$ kV) the reference voltages are power frequency voltage and lightning impulse voltage. In Range C ($U_m \geq 300$ kV) the reference voltage for external clearance is a switching impulse voltage. For further details see the introductions to Sub-clauses 3.2 and 3.3 of this standard. The conversion from rated withstand voltages to clearances in this standard is generally along the lines of IEC Publications 71-1 and 71-3.

3.1 Windings with $U_m < 300 \text{ kV}$

The same distance shall apply for clearances phase-to-earth, phase-to-neutral, phase-to-phase, and towards terminals of a lower voltage winding.

The recommended minimum clearances are given in Tables I and II with reference to the standard voltages which appear in Tables II and III of IEC Publication 76-3.

If a type test on a reduced clearance is to be conducted this shall be a lightning impulse test, dry, with positive impulse, three shots, with the test voltage according to Table I or II respectively.

3.2 Windings with $U_m \geq 300 \text{ kV}$ – Transformer specified according to Method 1

For equipment with U_m in this range, IEC Publication 71-3 specifies different requirements for phase-to-earth and phase-to-phase insulation. Air insulation clearances are referred to switching impulse conditions. Transformers which are specified according to Method 1 in IEC Publication 76-3 have, however, no rated switching impulse withstand level.

The recommended minimum distances in Table III have been chosen, directly on the basis of U_m values, from the selection of standard clearances appearing in Table IV. This is because it is assumed that the requirements for external insulation should be the same, irrespective of the use of Method 1 or Method 2 for the tests on the internal insulation.

The phase-to-earth clearances in Table III also apply for clearances towards the terminals of a lower voltage winding. (An unusual voltage ratio or connection may, however, require special consideration of the clearances.)

3.2.1 Type test procedure

If a type test on a reduced clearance is to be conducted this shall be carried out as described in Sub-clause 3.3.3. The test voltage is selected as follows:

- determine the standard clearance value;
- by reference to the appropriate curve of Figure 1 or 2, page 22, establish the switching impulse voltage corresponding to this standard clearance. This is the test voltage to be used.

3.3 Windings with $U_m \geq 300 \text{ kV}$ – Transformer specified according to Method 2

Transformers specified according to Method 2 have a rated switching impulse withstand voltage assigned to the highest voltage winding. This value, which applies between phase and earth, is also used as a reference for the external insulation.

The internal insulation is verified by a test with negative test voltage on the tested phase, and with 1.5 times test voltage between the phases on three-phase transformers.

For the external insulation the phase-to-phase withstand voltage is defined differently, according to IEC Publication 71-3 (see Table V). An appropriate test procedure involves positive polarity impulses for a configuration phase-to-earth, and opposite polarity impulses for phase-to-phase clearances (see Sub-clause 3.3.3). This has been considered for the clearance values given in Table IV.

3.3.1 *Clearance phase-to-earth, phase-to-neutral, and phase-to-phase between phases of the same winding*

The clearance from the high voltage bushing top to earth (tank, conservator, cooling equipment, switchyard structures) or to the neutral terminal is determined from column 4 of Table IV.

The clearance between bushing caps of different phases is determined from column 5 of Table IV.

3.3.2 *Clearance between terminals of different windings*

The clearance between terminals of different windings of the transformer shall be checked with regard to both switching impulse and lightning impulse conditions.

The switching impulse withstand requirement is based on the precalculated difference voltage which appears between the two terminals under consideration during the switching impulse test (see IEC Publication 76-3, Clause 14) according to the turns ratio between the windings. This voltage value gives the clearance which is required with regard to the switching impulse condition. Figure 2, page 22, is used to find the recommended clearance if the terminals receive opposite polarity voltages and the ratio between their voltage amplitudes is 2 or less. In other cases, Figure 1, page 22, applies.

Note. – If Figures 1 and 2 are compared it appears that a phase-to-phase clearance withstands a higher voltage difference than the same distance would do in a phase-to-earth configuration. The reason is that in the phase-to-phase configuration the two terminals are supposed to have opposite polarity, and the maximum dielectric gradient at either of them (which is largely determined by the voltage to earth) is relatively lower. It is assumed also that the electrodes have a rounded shape.

The clearance shall, however, also fulfil the lightning impulse withstand requirement, which presupposes that the lower voltage winding terminal is at earth potential when rated lightning impulse withstand voltage is applied to the high voltage terminal. The distance requirement in column 6 of Table IV and Figure 3, page 23, corresponding to this rated lightning impulse voltage, has therefore to be fulfilled between the two terminals.

The higher of the two clearance estimates shall apply.

The switching impulse test on the transformer will induce voltages between phases of other star-connected windings as well. It shall be checked whether this condition requires a larger phase-to-phase clearance in such a winding than as prescribed for this winding alone such as in Subclause 3.1.

3.3.3 *Type test procedure*

If a type test on a reduced clearance is to be conducted, the test procedure shall be as follows:

A test on a configuration phase-to-earth (or phase-to-neutral, or towards a terminal of a lower voltage winding) shall consist of a switching impulse test, dry, with positive polarity on the line terminal of the winding (the higher voltage winding). The counter electrode shall be earthed. If the tested terminal belongs to a three-phase winding, the other line terminals shall also be earthed.

Note. – This test is not generally feasible on complete three-phase transformers and may therefore have to be conducted on a model simulating the actual configuration of the transformer.

Tests on the phase-to-phase clearances of a three-phase transformer shall consist of switching impulse tests, dry, with half of the specified test voltage, positive, on one line terminal, the other half, negative, on another line terminal, and the third terminal earthed (see IEC Publication 71-3, Clauses 10, 12 and 14). The combinations of phase-to-earth and phase-to-phase test voltages,

according to IEC Publication 71-3, are reproduced in Table V. The values for $U_m = 525$ kV and 765 kV are subject to review.

When the outer phases are placed symmetrically with respect to the middle phase, it is sufficient to make two separate tests, one with positive polarity on the middle phase, and the other with positive polarity on an outer phase, the middle phase having negative polarity. If the line terminal arrangement is asymmetrical, it may be necessary to perform more than two tests.

Each test shall consist of 15 applications of impulse voltage with wave shape 250/2 500 in accordance with IEC Publication 60-2: High Voltage Test Techniques, Part 2: Test Procedures.

Note. – The above test procedure for phase-to-phase external clearances, which follows IEC Publication 71-3, differs in several respects from the switching impulse test procedure specified for the internal insulation of the transformer in Clause 14 of IEC Publication 76-3. The two test procedures do not replace each other.

TABLE I

*Recommended clearances from bushing live parts on power transformers
having windings with highest voltage for equipment $U_m < 300$ kV*

*Series I (based on current practice other than in the United States of America
and some other countries)*

Highest voltage for equipment U_m (r.m.s.) (kV)	Rated short duration power frequency withstand voltage (r.m.s.) (kV)	Rated lightning impulse withstand voltage (peak) (kV)		Minimum clearance (mm)	
		List 1	List 2	With reference to list 1	With reference to list 2
≤1.1	3	–	–	–	–
3.6	10	20	40	–	60
7.2	20	40	60	60	90
12	28	60	75	90	125
17.5	38	75	95	125	170
24	50	95	125	170	225
36	70	145	170	275	315
52	95		250		450
72.5	140		325		630
	185		450		830
123	230		550		1 050
145	275		650		1 250
170	325		750		1 450
	360		850		1 600
245	395		950		1 800

TABLE II

*Recommended clearances from bushing live parts on power transformers
having windings with highest voltage for equipment $U_m < 300 \text{ kV}$*

*Series II (based on current practice in the United States of America
and some other countries)*

Highest voltage for equipment U_m (r.m.s.) (kV)	Rated short duration power frequency withstand voltage (r.m.s.) (kV)	Rated lightning impulse withstand voltage (peak) (kV)		Minimum clearance (mm)	
		Distri- bution trans- formers	Other trans- formers	With reference to distribu- tion trans- formers	With reference to other trans- formers
4.40	19	60	75	65	100
13.20					
13.97	34	95	110	140	165
14.53					
26.4	50	150		225	
36.5	70	200		330	
72.5	140	350		630	
	185	450		830	
123					
	230	550		1050	
145					
	275	650		1250	
170					
	325	750		1450	
245					
	360	825		1600	
	395	900		1800	

TABLE III

*Recommended clearances from bushing live parts on power transformers
having windings with highest voltage for equipment $U_m \geq 300 \text{ kV}$,
specified according to Method 1*

Highest voltage for equipment U_m (r.m.s.) (kV)	Rated short duration power frequency withstand voltage (r.m.s.) (kV)	Rated lightning impulse withstand voltage (peak) (kV)	Minimum clearance (mm)	
			Phase-to earth	Phase-to phase
300	395	950	1900	2250
	460	1050	2300	2650
	460	1050	2300	2650
362	510	1175	2700	3100
	570	1300	2700	3100
420	630	1425	3100	3500

TABLE IV

Recommended clearances from bushing live parts on power transformers having windings with highest voltage for equipment $U_m \geq 300$ kV, specified according to Method 2

Highest voltage for equipment U_m (r.m.s.) (kV)	Rated switching impulse withstand voltage (peak) (kV)	Rated lightning impulse withstand voltage (peak) (kV)	Minimum clearance (mm)		
			Phase-to- earth, based on switching impulse with- stand voltage (see Sub- clause 3.3.1)	Phase-to phase, based on switching impulse with- stand voltage (see Sub- clause 3.3.1)	To other winding, based on lightning impulse with- stand voltage (see Sub- clause 3.3.2)
300	750	850	1900	2250	1600
		950			1750
362	850	1050	2300	2650	1950
		1175			2200
420	950	1300	2700	3100	2400
		1425			2650
525	1050	1550	3100	3500	2850
		1800			3300
765	1175	1950	3700	4200	3600
	1425		5000	5800	
	1550		5800	6700	

TABLE V

Combinations of switching impulse test voltages phase-to-earth and phase-to-phase for different values of U_m . The phase-to-phase voltage value for a type test according to Sub-clause 3.3.3 is selected according to the rated switching impulse withstand voltage, phase-to-earth, assigned to the transformer

Highest voltage for equipment U_m (r.m.s.) (kV)	Rated switching impulse withstand voltage phase-to-earth (peak) (kV)	Rated switching impulse withstand voltage phase-to-phase (peak) (kV)
300	750	1 175
	850	1 300
362	850	1 300
	950	1 425
420	950	1 425
	1 050	1 550
525	1 050	1 675
	1 175	1 800
765	1 425	2 400
	1 550	2 550

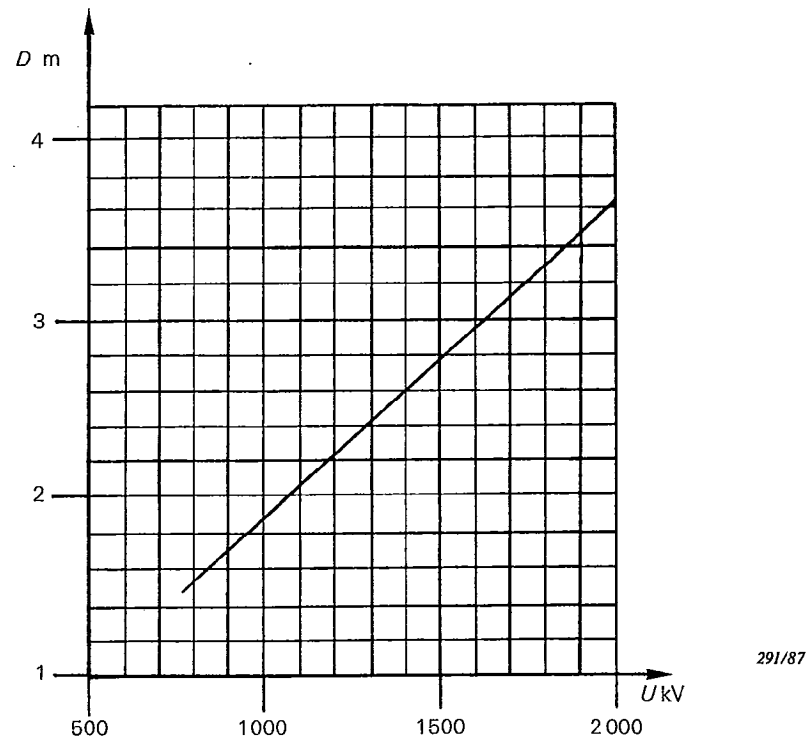


FIG. 3. - Distance d'isolement d'après la tension au choc de foudre.
Clearance based on lightning impulse voltage.